Financial Transaction Risk & Analytics

SQL + Power BI End-to-End Project Documentation

A professional transaction analytics dashboard for identifying high-value patterns, account/IP concentration, temporal behavior, and potential risk indicators.

Project component	Implementation
Database	MySQL
Database name	fraudanalytics
Main table	fraud_transactions
BI / Visualization	Microsoft Power BI
Dataset size	8,430 transactions (shown as 8.43K)
Dashboard pages	Transaction Overview • Transaction Risk Analysis • Transaction Investigation

Important scope note: The available transaction table does not contain a confirmed fraud label. Therefore, this project should be presented as transaction risk analytics and investigation support—not as a model that confirms fraudulent transactions.

1. Executive Summary

The project analyzes financial transaction records using MySQL for data preparation/validation and Power BI for interactive analysis. The dashboard is designed as a three-stage analytical flow: **Overview** → **Risk Analysis** → **Investigation**.

• **Overview:** establishes transaction volume, total value, currency distribution, hourly activity, IP concentration, and time trends.

• **Risk Analysis:** focuses on average transaction amounts across currencies, hours, accounts, IP addresses, and dates.

• **Investigation:** enables filtering and drill-style review of high-value transactions, accounts, IPs, and transaction-level details.

Business objective

Help an analyst quickly identify unusual concentration or high-value behavior that may deserve further investigation. The dashboard is descriptive and diagnostic; it does not independently determine whether a transaction is fraudulent.

Tools used

Tool	Role in the project
MySQL	Database storage, SQL querying, validation, and transaction-level analysis.
Power Query	Data-type checks, cleaning, transformation, and preparation inside Power BI.
Power BI	Interactive dashboard design, slicers, aggregations, filtering, and visual analytics.

2. Dataset & Data Quality

The primary Power BI table used in the final dashboard is **fraud_transactions** from the **fraudanalytics** MySQL database. The final dashboard contains approximately 8,430 transaction records.

Fields used

Field	Purpose
accountID	Identify and compare transaction activity across accounts.
localHour	Analyze transaction behavior by hour of day (0–23).
transactionAmount	Core monetary measure for totals, averages, and high-value analysis.
transactionCurrencyCode	Compare transaction activity and amounts by currency.
transactionDate	Original transaction date field.
TransactionDate_Clean	Cleaned date field used for time-based Power BI analysis.
transactionID	Unique transaction-level identifier used in investigation tables/charts.
transactionIPAddress	Analyze transaction concentration by IP address.
transactionTime	Transaction time value used in the investigation table.

Data-quality decisions

• The cleaned date field **TransactionDate_Clean** was used for reliable date-based visuals.

• transactionDeviceId was investigated for quality. It contained only the value **NA** as a distinct value, so it was removed rather than used as a misleading analytical dimension.

• The separate Excel profiling dataset containing fraud-related fields was not mixed with this transaction dataset, preventing inconsistent analysis across different sources.

• IP values are treated as the dataset provides them; they are not assumed to be conventional dotted IPv4 strings.

Result: the final model is intentionally limited to fields that contain usable analytical information.

3. SQL / MySQL Stage

MySQL was used as the structured data layer before the Power BI reporting stage. The workflow focused on understanding the transaction table, validating fields, and preparing the dataset for BI analysis.

SQL workflow

Step	Purpose	Output
1. Database setup	Work inside fraudanalytics and access fraud_transactions.	Usable transaction table
2. Schema inspection	Review columns and data types.	Understanding of available dimensions/measures
3. Data validation	Check dates, amounts, identifiers, and categorical fields.	Cleaner analytical base
4. Exploratory analysis	Review transaction counts, amounts, currencies, hours, accounts, and IPs.	Patterns to visualize
5. Power BI handoff	Use the validated transaction table for interactive reporting.	BI-ready dataset

Core analytical logic used in the dashboard

Metric / Analysis	Logic
Total Transactions	Count of transactionID
Total Transaction Amount	Sum of transactionAmount
Average Transaction Amount	Average of transactionAmount
Transactions by Currency	Count of transactionID grouped by transactionCurrencyCode
Transactions by Hour	Count of transactionID grouped by localHour
Average Amount by Currency	Average transactionAmount grouped by transactionCurrencyCode
Average Amount by Hour	Average transactionAmount grouped by localHour
Top Accounts	Top 10 accounts ranked by average or total transaction amount, depending on page
Top IPs	Top 10 IP addresses ranked by transaction count or transaction amount, depending on page

4. Power BI Data Preparation & Model

After the MySQL stage, the transaction table was brought into Power BI and checked through Power Query. The dashboard uses a single primary transaction table, which keeps the model simple and appropriate for this dataset.

Preparation steps

• Verify field types and validity before building visuals.

• Use TransactionDate_Clean for date-based analysis.

• Remove the non-informative device field after quality checking.

• Use transactionAmount as the main numerical measure.

• Use accountID, transactionCurrencyCode, localHour, transactionIPAddress, and transactionID as analytical dimensions.

Important Power BI design choices

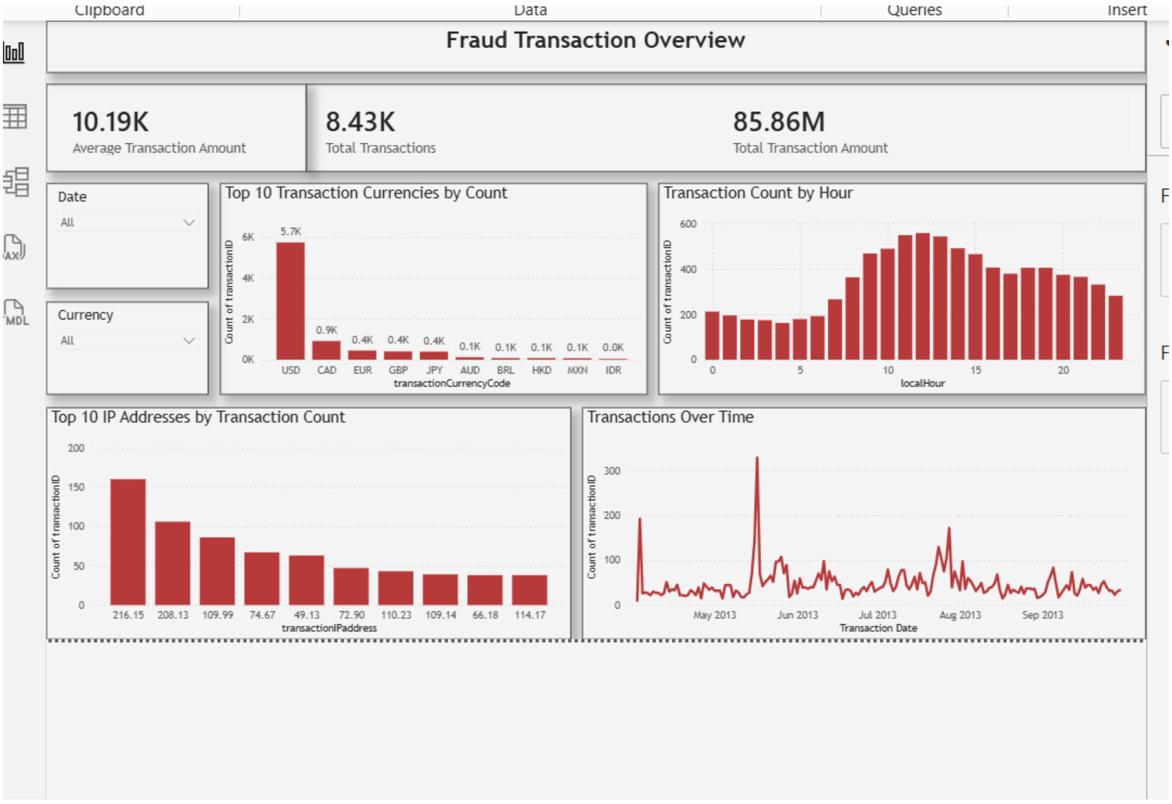
Design choice	Reason
Dropdown slicers	Compact and easy to use while preserving dashboard space.
Top 10 filters	Focus attention on the most concentrated accounts/IPs without overwhelming the page.
Average aggregation	Used on Risk Analysis to study high-value behavior rather than only transaction frequency.
Sum aggregation	Used on Investigation where total monetary exposure is the relevant measure.
Line charts	Used for time-series behavior where continuity across dates matters.
Treemap	Used on Investigation to show relative total transaction amount across top accounts.
Scatter chart	Used during IP analysis to highlight monetary concentration by IP.

5. Dashboard Page 1 — Transaction Overview

Purpose: provide a fast, high-level view of transaction volume, monetary value, currency mix, hourly activity, IP concentration, and transaction trends.

Key elements

- KPI — Average Transaction Amount: **10.19K**
- KPI — Total Transactions: **8.43K**
- KPI — Total Transaction Amount: **85.86M**
- Slicers — Date and Currency
- Top 10 Transaction Currencies by Count
- Transaction Count by Hour
- Top 10 IP Addresses by Transaction Count
- Transactions Over Time



Analytical purpose

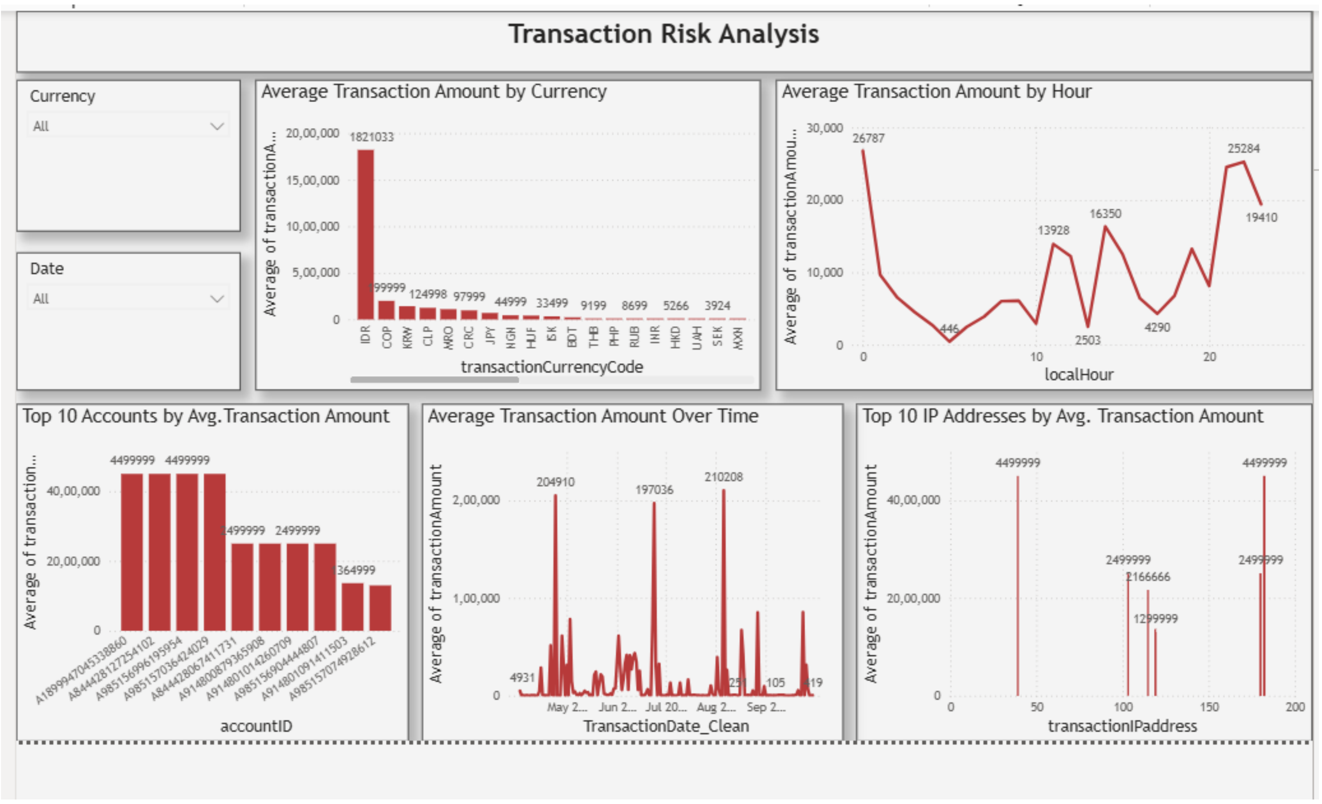
This page answers: *How much activity is occurring, when is it occurring, which currencies dominate, and where is transaction activity concentrated?*

6. Dashboard Page 2 — Transaction Risk Analysis

Purpose: move from transaction volume to monetary behavior. The page emphasizes average transaction amount and highlights areas where higher-value activity is concentrated.

Key elements

- Slicers — Currency and Date
- Average Transaction Amount by Currency
- Average Transaction Amount by Hour
- Top 10 Accounts by Avg. Transaction Amount
- Average Transaction Amount Over Time
- Top 10 IP Addresses by Avg. Transaction Amount



Analytical purpose

This page answers: *Which currencies, hours, accounts, IPs, and dates are associated with higher average transaction values?* This is a risk-indicator view, not a fraud-confirmation view.

7. Dashboard Page 3 — Transaction Investigation

Purpose: provide an investigation-oriented workspace where an analyst can filter the dataset and inspect high-value transactions, accounts, IP addresses, and individual transaction records.

Filters

• Currency

• Account

• Transaction Date

Visuals

Visual	Purpose
Top 10 Highest-Value Transactions	Identify individual transactions with the largest monetary values.
Top 10 Accounts by Total Transaction Amount	Show accounts with the greatest total monetary exposure.
Top 10 IP Addresses by Transaction Amount	Identify IP addresses associated with high total transaction value.
Transaction Details table	Review transactionID, accountID, date, time, amount, currency, IP, and hour.

Investigation workflow

• Start with the highest-value transactions.

• Use the account/IP visuals to identify concentration.

• Apply Currency, Account, or Date filters to narrow the investigation.

• Use the transaction table for transaction-level review.

Important: a high-value or concentrated transaction is a potential review signal, not proof of fraud.

8. Key Insights & Business Interpretation

The dashboard is designed to support the following types of findings:

Insight area	What the dashboard can reveal	Business use
Currency concentration	A small number of currencies can account for most transaction activity.	Prioritize monitoring where activity is concentrated.
Hourly behavior	Transaction frequency and average value vary by hour.	Identify unusually active or high-value time windows.
Account concentration	A small group of accounts may contribute disproportionately to transaction value.	Focus account-level review on high-exposure entities.
IP concentration	A small group of IP values can account for substantial transaction volume.	Investigate unusual concentration or shared access pat
Time spikes	Certain dates can show sharp increases in transaction amount or count.	Review event-driven or anomalous periods.
Transaction-level outliers	A few transactions can be much larger than the typical transaction.	Escalate high-value records for additional validation.

Example interview interpretation

“I used the first page to understand overall transaction behavior, the second page to identify higher-value risk patterns, and the third page to investigate individual transactions and concentration by account and IP. Since the source data did not contain a confirmed fraud label, I positioned the output as transaction risk and investigation analytics rather than a fraud-classification model.”

9. Limitations, Assumptions & Interview Notes

Limitations

• No fraud_flag / confirmed fraud outcome is present in the final transaction table.

• Therefore, the dashboard cannot calculate fraud precision, recall, accuracy, or confirmed fraud rates.

• No merchant category, transaction type, geographic location, or usable device identifier is available in the final table.

• The analysis is descriptive: it identifies patterns and potential risk indicators that require further investigation.

Recommended future enhancements

• Add a validated fraud label from a trusted source.

• Add merchant category and transaction type.

• Add geography and customer/account profile information where appropriate.

• Add validated device identifiers and device-change behavior.

• Create a supervised fraud-risk model once labeled historical data is available.

30-second interview pitch

“I built a financial transaction analytics dashboard using MySQL and Power BI. I first validated and explored the transaction data in SQL, then prepared the data in Power Query and created a three-page Power BI dashboard. The first page covers transaction overview, the second focuses on high-value risk patterns, and the third supports transaction investigation using account, IP, date, and transaction-level analysis. The key technical focus was choosing the right aggregation—count for activity, average for risk-pattern analysis, and total amount for investigation.”

Final project positioning

Recommended project title: Financial Transaction Risk & Analytics Dashboard using SQL and Power BI

Recommended resume description: Built an interactive Power BI dashboard using MySQL transaction data to analyze transaction volume, high-value behavior, account/IP concentration, hourly patterns, and transaction-level investigation.

Appendix — Complete SQL Queries

The first version of the documentation described the SQL workflow but did not include the actual SQL statements. This appendix fixes that omission and records the MySQL queries used during the transaction-analysis workflow.

Database: fraudanalytics **Main table:** fraud_transactions

Note: The queries are transcribed from the SQL work and screen captures available for this project. The final Power BI dashboard uses Power Query for the cleaned date field and Power BI aggregations for its visuals.

1. Create and select the database

```
CREATE DATABASE FraudAnalytics;
USE FraudAnalytics;
```

2. Create the transaction table

```
CREATE TABLE transactions (
    transactionID VARCHAR(50),
    accountID VARCHAR(52),
    transactionAmount DECIMAL(15,2),
    transactionCurrencyCode VARCHAR(12),
    transactionDate DATE,
    transactionTime TIME,
    localHour INT,
    transactionDeviceId VARCHAR(100),
    transactionIPAddress VARCHAR(50)
);
```

3. Inspect the available tables

```
SHOW TABLES;
```

4. Inspect the transaction table structure

```
DESCRIBE fraud_transactions;
```

5. Count total transaction records

```
SELECT COUNT(*) AS total_transactions
FROM fraud_transactions;
```

6. Inspect a sample of transactions

```
SELECT *
FROM fraud_transactions
LIMIT 10;
```

7. Check the transaction date range

```
SELECT
    MIN(transactionDate) AS first_date,
    MAX(transactionDate) AS last_date
FROM fraud_transactions;
```

8. Check completeness of key fields

```
SELECT
    COUNT(*) AS total_rows,
    COUNT(transactionID) AS transaction_ids,
    COUNT(accountID) AS account_ids,
    COUNT(transactionAmount) AS amounts,
    COUNT(transactionDeviceId) AS devices,
    COUNT(transactionIPAddress) AS ip_addresses
FROM fraud_transactions;
```

9. Check duplicate transaction IDs

```
SELECT
    transactionID,
    COUNT(*) AS count
FROM fraud_transactions
GROUP BY transactionID
```

```
HAVING COUNT(*) > 1;
```

10. Check transaction amount statistics

```
SELECT
    MIN(transactionAmount) AS minimum_amount,
    MAX(transactionAmount) AS maximum_amount,
    AVG(transactionAmount) AS average_amount
FROM fraud_transactions;
```

11. Identify high-value transactions

```
SELECT
    transactionID,
    accountID,
    transactionAmount
FROM fraud_transactions
WHERE transactionAmount > 100000
ORDER BY transactionAmount DESC;
```

12. Analyze transactions and average amount by hour

```
SELECT
    localHour,
    COUNT(*) AS total_transactions,
    AVG(transactionAmount) AS average_amount
FROM fraud_transactions
GROUP BY localHour
ORDER BY average_amount DESC;
```

13. Analyze transaction count by currency

```
SELECT
    transactionCurrencyCode,
    COUNT(*) AS transactions
FROM fraud_transactions
GROUP BY transactionCurrencyCode
ORDER BY transactions DESC;
```

14. Check transaction distribution by device

```
SELECT
    transactionDeviceId,
    COUNT(*) AS transactions
FROM fraud_transactions
GROUP BY transactionDeviceId
ORDER BY transactions DESC;
```

15. Analyze top IP addresses by transaction count

```
SELECT
    transactionIPAddress,
    COUNT(*) AS transactions
FROM fraud_transactions
GROUP BY transactionIPAddress
ORDER BY transactions DESC
LIMIT 10;
```

16. Analyze account activity

```
SELECT
    accountID,
    COUNT(*) AS transactions
FROM fraud_transactions
GROUP BY accountID
ORDER BY transactions DESC
LIMIT 10;
```

17. Identify accounts with highest total transaction amount

```
SELECT
    accountID,
    SUM(transactionAmount) AS total_amount
FROM fraud_transactions
GROUP BY accountID
ORDER BY total_amount DESC
```

```
LIMIT 10;
```

18. Find accounts with very high total transaction amount

```
SELECT
    accountID,
    SUM(transactionAmount) AS total_amount
FROM fraud_transactions
GROUP BY accountID
HAVING SUM(transactionAmount) > 1000000;
```

19. Find high-value transaction activity by hour

```
SELECT
    localHour,
    COUNT(*) AS transactions
FROM fraud_transactions
WHERE transactionAmount > 100000
GROUP BY localHour
ORDER BY transactions DESC;
```

20. Final transaction summary

```
SELECT
    COUNT(*) AS transactions,
    SUM(transactionAmount) AS total_amount,
    AVG(transactionAmount) AS average_amount
FROM fraud_transactions;
```

SQL-to-Power BI Mapping

Dashboard requirement	SQL query / logic
Total Transactions	Query 5 / COUNT(transactionID)
Total Transaction Amount	Query 20 / SUM(transactionAmount)
Average Transaction Amount	Query 20 / AVG(transactionAmount)
Top currencies by count	Query 13
Transaction count by hour	Query 12
Top IPs by transaction count	Query 15
High-value transactions	Query 11
Top accounts by total amount	Query 17
High-value activity by hour	Query 19
Risk Analysis averages	AVG(transactionAmount) grouped by currency/hour/account/IP
Investigation table	Transaction-level fields from fraud_transactions

Scope reminder: No confirmed fraud label is present in the final transaction table. The SQL and Power BI outputs identify transaction patterns, high-value behavior and potential risk indicators; they do not confirm fraud.